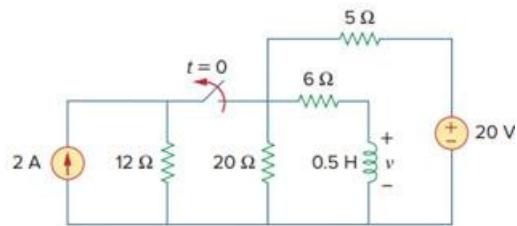


Problem

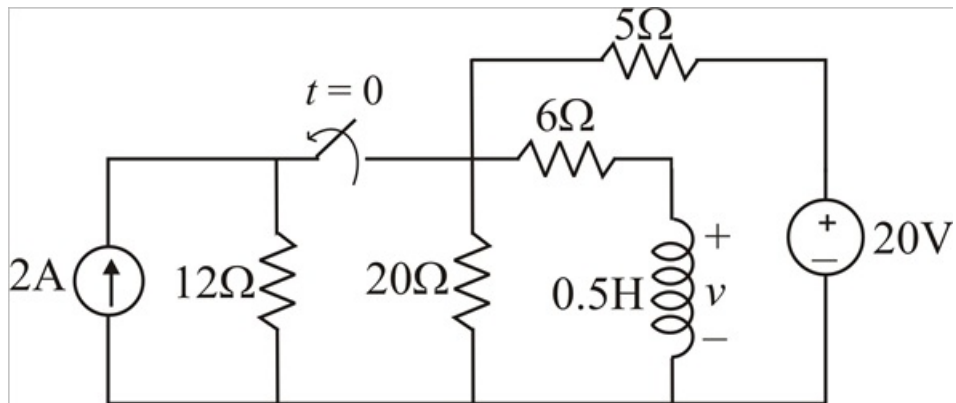
For the network shown in Fig. 7.122, find $v(t)$ for $t > 0$.



Step-by-step solution

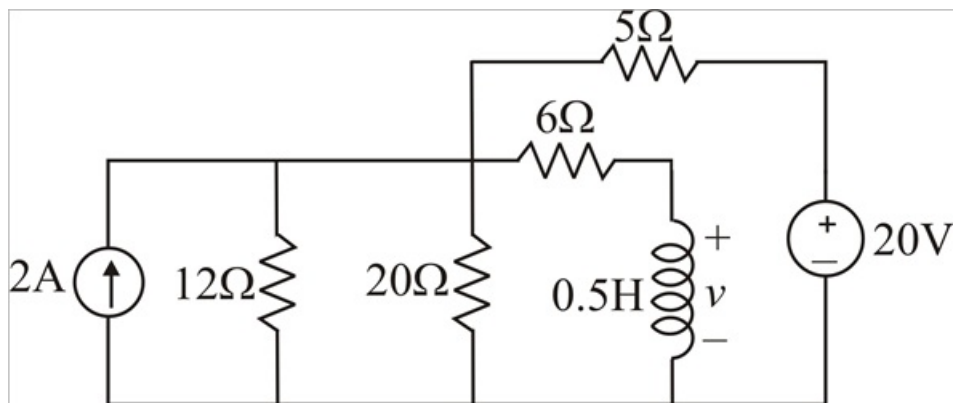
Step 1 of 12

Given circuit diagram



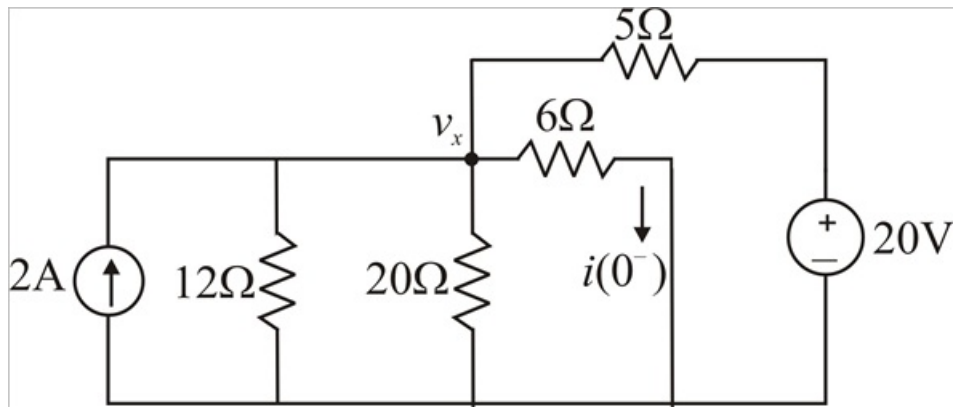
Step 2 of 12

For $t < 0$ switch is on closed position so the circuit is as below



Step 3 of 12

For dc inductor acts like a short circuit, given circuit is as below at $t = 0^-$



Step 4 of 12

By applying KCL at node v_x we can write

$$2 = \frac{v_x - 20}{5} + \frac{v_x}{12} + \frac{v_x}{20} + \frac{v_x}{6}$$

$$120 = 12v_x - 240 + 5v_x + 3v_x + 10v_x$$

$$30v_x = 360$$

$$v_x = 12V$$

Step 5 of 12

From the above circuit

$$i(0^-) = \frac{v_x}{6}$$

$$i(0^-) = \frac{12}{6}$$

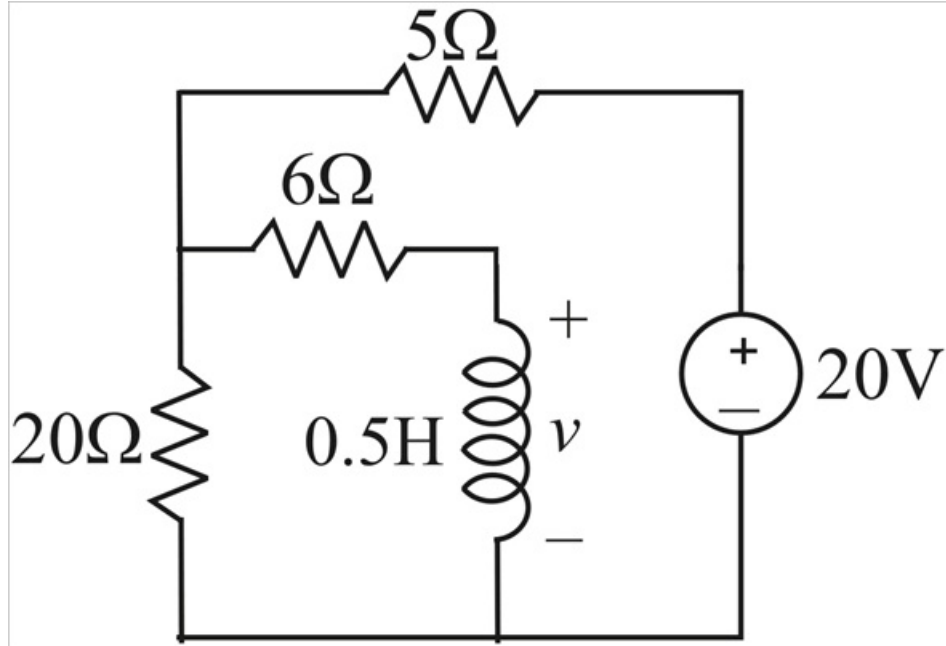
$$i(0^-) = 2A$$

Step 6 of 12

Inductor current can not change instantaneously

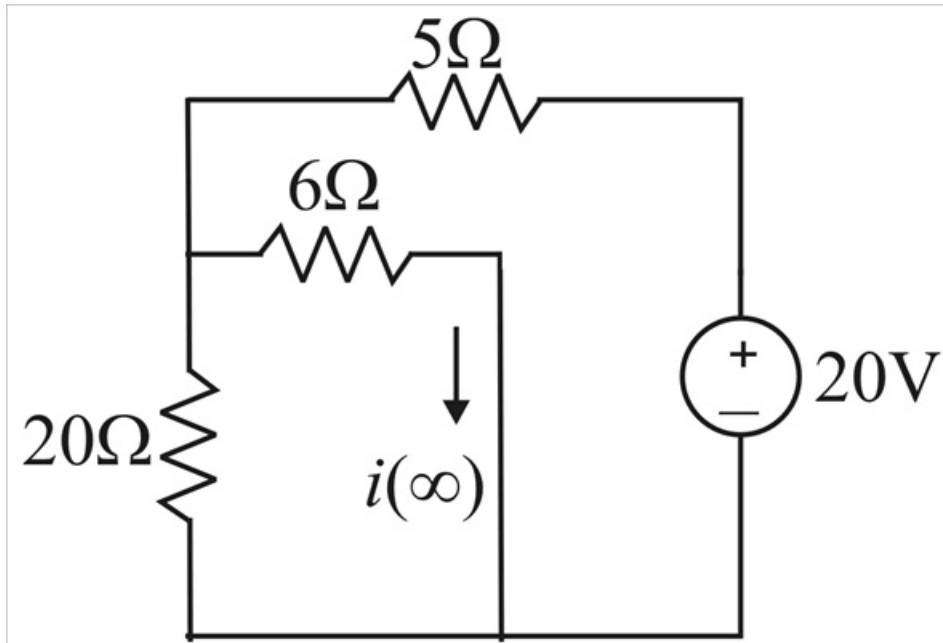
$$i(0^-) = i(0^+) = i(0) = 2\text{A}$$

For $t > 0$ given circuit is as below



Step 7 of 12

In the above circuit for dc inductor acts like short circuit and at $t = \infty$ given circuit is as below



From the above circuit current flowing through 20V source is

$$I_{20} = \frac{20}{5 + (6 \parallel 20)}$$

$$I_{20} = \frac{20}{5 + \left(\frac{6 \times 20}{6 + 20}\right)}$$

$$I_{20} = 2.08\text{A}$$

By using current division rule we can write

$$i(\infty) = I_{20} \left(\frac{20}{20 + 6} \right)$$

$$i(\infty) = (2.08) \left(\frac{20}{26} \right)$$

$$i(\infty) = 1.6\text{A}$$

Step 8 of 12

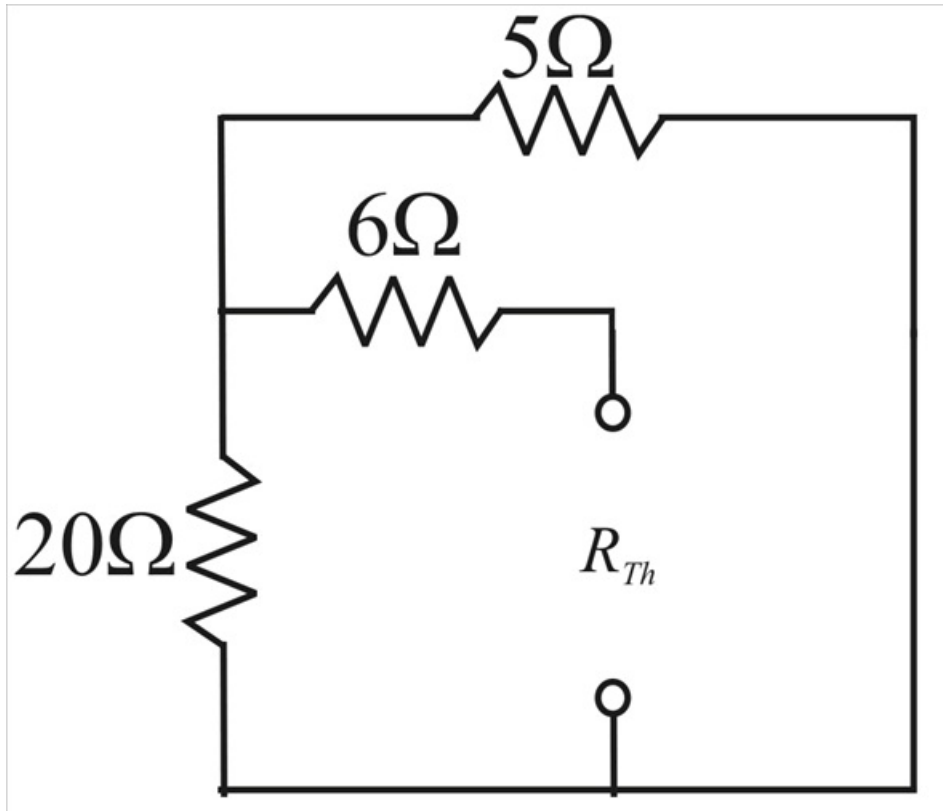
Time constant of the circuit for $t > 0$

$$\tau = \frac{L}{R_{th}}$$

Here R_{th} is Thevenin's equivalent resistance seen across inductor terminals by replacing 20V with short circuit

Step 9 of 12

R_{Th} is determined from the below circuit



Step 10 of 12

$$R_{Th} = 6 + (20 \parallel 5)$$

$$R_{Th} = 6 + \left(\frac{20 \times 5}{20 + 5} \right)$$

$$R_{Th} = 6 + 4$$

$$R_{Th} = 10\Omega$$

$$\tau = \frac{0.5}{10}$$

$$\tau = 0.05 \text{ sec}$$

Step 11 of 12

General equation for inductor current is

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

$$i(t) = 1.6 + [2 - 1.6]e^{-\frac{t}{0.05}}$$

$$i(t) = (1.6 + 0.4e^{-20t}) \text{ A}$$

Step 12 of 12

Inductor voltage

$$v(t) = L \frac{di(t)}{dt}$$

$$v(t) = (0.5) \frac{d}{dt} (1.6 + 0.4e^{-20t})$$

$$v(t) = (0.5) [0.4(-20)e^{-20t}]$$

$$v(t) = -4e^{-20t} \text{ V}$$

[Comments \(1\)](#)



Anonymous

V(t)=0 since inductor would be shorted from DC voltage source
